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-- USERS TABLE

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CREATE TABLE IF NOT EXISTS users (
    user_id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    surname VARCHAR(100) NOT NULL,
    date_created TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

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-- USER ACCOUNT TABLE (1:1 with users)

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CREATE TABLE IF NOT EXISTS user_account (
    account_id SERIAL PRIMARY KEY,
    user_id INT UNIQUE NOT NULL REFERENCES users(user_id) ON DELETE CASCADE,
    account_num VARCHAR(50) UNIQUE NOT NULL,
    amount NUMERIC(12,2) DEFAULT 0
);

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-- GOAL TABLE

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CREATE TABLE IF NOT EXISTS goal (
    goal_id SERIAL PRIMARY KEY,
    goal_name VARCHAR(255) NOT NULL,

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goal_amount NUMERIC(12,2) NOT NULL,
goal_term VARCHAR(100),
total_transact NUMERIC(12,2) DEFAULT 0,
remaining_amount NUMERIC(12,2) DEFAULT 0,
status VARCHAR(20) NOT NULL DEFAULT 'active'
    CHECK (status IN ('active','matured','closed'))
);

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-- USER TRANSACTIONS TABLE

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CREATE TABLE IF NOT EXISTS user_transact (
    transact_id SERIAL PRIMARY KEY,
    account_id INT NOT NULL REFERENCES user_account(account_id),
    amount_transacted NUMERIC(12,2) NOT NULL,
    transaction_type VARCHAR(20) NOT NULL
    CHECK (transaction_type IN ('deposit','withdraw','transfer')),
    datetime TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    goal_id INT REFERENCES goal(goal_id)
);

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-- TRIGGERS

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-- Prevent withdrawals until goal is matured

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CREATE OR REPLACE FUNCTION prevent_withdrawals()
RETURNS TRIGGER AS $$
BEGIN
    IF NEW.transaction_type = 'withdraw' AND NEW.goal_id IS NOT NULL THEN
        IF (SELECT status FROM goal WHERE goal_id = NEW.goal_id) <> 'matured' THEN
            RAISE EXCEPTION 'Withdrawals not allowed until goal is matured.';
        END IF;
    END IF;
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;
```

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CREATE TRIGGER trg_prevent_withdrawals
BEFORE INSERT ON user_transact
FOR EACH ROW
EXECUTE FUNCTION prevent_withdrawals();
```

-- Auto-update goal totals on deposits

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CREATE OR REPLACE FUNCTION update_goal_totals()
RETURNS TRIGGER AS $$
BEGIN
    IF NEW.transaction_type = 'deposit' AND NEW.goal_id IS NOT NULL THEN
        UPDATE goal
        SET total_transact = total_transact + NEW.amount_transacted,
            remaining_amount = goal_amount - (total_transact + NEW.amount_transacted)
        WHERE goal_id = NEW.goal_id;
```

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UPDATE goal
SET status = 'matured'
WHERE goal_id = NEW.goal_id AND total_transact >= goal_amount;
END IF;
RETURN NEW;
END;
$$ LANGUAGE plpgsql;
```

```
CREATE TRIGGER trg_update_goal_totals
AFTER INSERT ON user_transact
FOR EACH ROW
EXECUTE FUNCTION update_goal_totals();
```